
Unit III – Integration

Introduction

Calculus is the mathematical study of continuous change.

Calculus – Differentiation and integration.

- Differential Calculus is centred on the concept of the derivative. Differentiation is the process of finding slope of a curve. Slope is the change in y divided by change in x .
- Integral Calculus is used to find the area between graph of a curve and x -axis. It is the process of cutting up an area into small sections and then adding up areas to get total area. Integration means summation. The integral calculus is the study of integration of functions. This finds extensive applications in Geometry, Mechanics, Natural sciences and other disciplines.

Integration as inverse of differentiation

Examples:

$$1. \frac{d}{dx}(x^2) = 2x, \frac{d}{dx}(x^2 + c) = 2x$$

x^2 is the integral of $2x$ w.r.t. x and is written as $\int 2x dx = x^2$

$$2. \frac{d}{dx}(\sin x) = \cos x, \frac{d}{dx}(\sin x + c) = \cos x$$

$$3. \frac{d}{dx}(e^x) = e^x, \frac{d}{dx}(e^x + c) = e^x$$

Definition: If $f(x)$ and $g(x)$ are two functions such that $\frac{d}{dx}[f(x)] = f'(x) = g(x)$ then $f(x)$ is said to be an integral (antiderivative or primitive) of $g(x)$ with respect to x (w.r.t. x) and is written $\int g(x) dx = f(x)$. The process of finding the primitive or integral of a function is called integration.

Note:

- The symbol $\int$ is called the integral sign.
- In $\int g(x) dx = f(x)$, $g(x)$ is an integrand and x is the variable of integration.

-
- $\frac{d}{dx}[f(x)] = g(x)$ and also $\frac{d}{dx}[f(x) + c] = g(x)$, therefore $\int g(x)dx = f(x) + c$, where c is any arbitrary constant and is called the constant of integration.

- $\int g(x)dx = f(x) + c$ is called the “indefinite integral”

Note: $\frac{d}{dx}[f(x)] = g(x)$. Integrating both sides, we get $\int \frac{d}{dx}[f(x)] dx = \int g(x) dx$ i.e., $\int g(x)dx = f(x) + c$

Symbols/Terms/Phrases	Meaning
$\int f(x) dx$	Integral of f with respect to x
$f(x)$ in $\int f(x) dx$	Integrand
x in $\int f(x) dx$	Variable of integration
Integrate	Find the integral
An integral of f	A function F such that $F'(x) = f(x)$
Integration	The process of finding the integral
Constant of Integration	Any real number C , considered as constant function

Trigonometric formulaes

$$\sin(-\theta) = -\sin(\theta)$$

$$\cos(-\theta) = \cos(\theta)$$

$$\sin(A + B) = \sin(A) \cos(B) + \cos(A) \sin(B)$$

$$\cos(A + B) = \cos(A) \cos(B) - \sin(A) \sin(B)$$

$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$$

$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$$

$$\sin(A) \sin(B) = \frac{\cos(A - B) - \cos(A + B)}{2}$$

$$\cos(A) \cos(B) = \frac{\cos(A + B) + \cos(A - B)}{2}$$

$$\sin(A) \cos(B) = \frac{\sin(A + B) + \sin(A - B)}{2}$$

$$\cos(A) \sin(B) = \frac{\sin(A + B) - \sin(A - B)}{2}$$

$$\sin(A) + \sin(B) = 2 \sin\left(\frac{A + B}{2}\right) \cos\left(\frac{A - B}{2}\right)$$

$$\cos(A) + \cos(B) = -2 \cos\left(\frac{A + B}{2}\right) \cos\left(\frac{A - B}{2}\right)$$

Integral of standard functions

(i) $\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = x^n, n \neq -1$	$\Rightarrow \int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
(ii) $\frac{d}{dx} (\log_e x) = \frac{1}{x}$	$\Rightarrow \int \frac{1}{x} dx = \log_e x + C$
(iii) $\frac{d}{dx} (e^x) = e^x$	$\Rightarrow \int e^x dx = e^x + C$
(iv) $\frac{d}{dx} \left(\frac{a^x}{\log_e a} \right) = a^x, a > 0, a \neq 1$	$\Rightarrow \int a^x dx = \frac{a^x}{\log_e a} + C$
(v) $\frac{d}{dx} (-\cos x) = \sin x$	$\Rightarrow \int \sin x dx = -\cos x + C$
(vi) $\frac{d}{dx} (\sin x) = \cos x$	$\Rightarrow \int \cos x dx = \sin x + C$
(vii) $\frac{d}{dx} (\tan x) = \sec^2 x$	$\Rightarrow \int \sec^2 x dx = \tan x + C$
(viii) $\frac{d}{dx} (-\cot x) = \operatorname{cosec}^2 x$	$\Rightarrow \int \operatorname{cosec}^2 x dx = -\cot x + C$
(ix) $\frac{d}{dx} (\sec x) = \sec x \tan x$	$\Rightarrow \int \sec x \tan x dx = \sec x + C$
(x) $\frac{d}{dx} (-\operatorname{cosec} x) = \operatorname{cosec} x \cot x$	$\Rightarrow \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$
(xi) $\frac{d}{dx} (\log \sin x) = \cot x$	$\Rightarrow \int \cot x dx = \log \sin x + C$
(xii) $\frac{d}{dx} (-\log \cos x) = \tan x$	$\Rightarrow \int \tan x dx = -\log \cos x + C$
(xiii) $\frac{d}{dx} \{\log (\sec x + \tan x)\} = \sec x$	$\Rightarrow \int \sec x dx = \log \sec x + \tan x + C$
(xiv) $\frac{d}{dx} \{\log (\operatorname{cosec} x - \cot x)\} = \operatorname{cosec} x$	$\Rightarrow \int \operatorname{cosec} x dx = \log \operatorname{cosec} x - \cot x + C$
(xv) $\frac{d}{dx} \left(\sin^{-1} \frac{x}{a} \right) = \frac{1}{\sqrt{a^2 - x^2}}$	$\Rightarrow \int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + C$
(xvi) $\frac{d}{dx} \left(\cos^{-1} \frac{x}{a} \right) = -\frac{1}{\sqrt{a^2 - x^2}}$	$\Rightarrow \int -\frac{1}{\sqrt{a^2 - x^2}} dx = \cos^{-1} \left(\frac{x}{a} \right) + C$
(xvii) $\frac{d}{dx} \left(\frac{1}{a} \tan^{-1} \frac{x}{a} \right) = \frac{1}{a^2 + x^2}$	$\Rightarrow \int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$
(xviii) $\frac{d}{dx} \left(\frac{1}{a} \cot^{-1} \frac{x}{a} \right) = -\frac{1}{a^2 + x^2}$	$\Rightarrow \int -\frac{1}{a^2 + x^2} dx = \frac{1}{a} \cot^{-1} \left(\frac{x}{a} \right) + C$
(xix) $\frac{d}{dx} \left(\frac{1}{a} \sec^{-1} \frac{x}{a} \right) = \frac{1}{x \sqrt{x^2 - a^2}}$	$\Rightarrow \int \frac{1}{x \sqrt{x^2 - a^2}} dx = \frac{1}{a} \sec^{-1} \left(\frac{x}{a} \right) + C$
(xx) $\frac{d}{dx} \left(\frac{1}{a} \operatorname{cosec}^{-1} \frac{x}{a} \right) = -\frac{1}{x \sqrt{x^2 - a^2}}$	$\Rightarrow \int -\frac{1}{x \sqrt{x^2 - a^2}} dx = \frac{1}{a} \operatorname{cosec}^{-1} \left(\frac{x}{a} \right) + C$

Properties of Integrals

Let $f(x)$ and $g(x)$ be two functions, then

- $\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx$
- $\int [f(x) - g(x)]dx = \int f(x)dx - \int g(x)dx$
- $\int kf(x)dx = k \int f(x)dx$

Note:

1. $\int \frac{1}{f(x)} dx$ will be often written as $\int \frac{dx}{f(x)}$
2. If $\int f(x)dx = F(x) + c$, then
 - a. $\int f(ax)dx = \frac{F(ax)}{a} + c$
 - b. $\int f(ax + b)dx = \frac{F(ax+b)}{a} + c$

Examples:

$$\int \cos(3x + 4) dx = \frac{\sin(3x+4)}{3}, \int e^{4x+5} dx = \frac{e^{4x+5}}{4}$$

Problems

Evaluate the following integrals

1. $\int x^3 dx$

Solution:

We know $\int x^n dx = \frac{x^{n+1}}{n+1} + c$

$$\therefore \int x^3 dx = \frac{x^{3+1}}{3+1} = \frac{x^4}{4} + c$$

2. $\int 10x dx$

Solution:

We have $\int 10x dx = 10 \int x dx = 10 \cdot \frac{x^{1+1}}{1+1} + c$

$$= 10 \cdot \frac{x^2}{2} + c = 5x^2 + c$$

3. $\int x^{\frac{2}{3}} dx$

Solution:

We know $\int x^n dx = \frac{x^{n+1}}{n+1} + c$

$$\therefore \int x^{\frac{2}{3}} dx = \frac{x^{\frac{2}{3}+1}}{\frac{2}{3}+1} + c = \frac{x^{\frac{5}{3}}}{\frac{5}{3}} + c = \frac{3x^{\frac{5}{3}}}{5} + c$$

4. $\int \frac{1}{x^5} dx$

Solution:

We know $\int x^n dx = \frac{x^{n+1}}{n+1} + c$

$$\therefore \int \frac{1}{x^5} dx = \int x^{-5} dx = \frac{x^{-5+1}}{-5+1} = \frac{x^{-4}}{-4} + c$$

5. $\int \left(1 + \frac{1}{x}\right) dx$

Solution:

We know $\int \left(1 + \frac{1}{x}\right) dx = \int 1 \cdot dx + \int \frac{1}{x} dx = \int x^0 \cdot dx + \int \frac{1}{x} dx$

$$= \frac{x^{0+1}}{0+1} + \log x + c$$

$$\frac{x^1}{1} + \log x + c = x + \log x + c$$

6. $\int (5x^2 - 8x + 5) dx$

Solution:

$$\int (5x^2 - 8x + 5) dx = \int 5x^2 dx - \int 8x dx + \int 5 dx$$

$$= 5 \int x^2 dx - 8 \int x dx + 5 \int dx$$

$$= 5 \cdot \frac{x^{2+1}}{2+1} - 8 \cdot \frac{x^{1+1}}{1+1} + 5 \cdot \frac{x^{0+1}}{0+1} + c$$

$$= 5 \cdot \frac{x^3}{3} - 8 \cdot \frac{x^2}{2} + 5 \cdot \frac{x^1}{1} + c$$

$$= \frac{5x^3}{3} - 4x^2 + 5x + c$$

7. $\int \left(\sqrt{x} + \frac{1}{3\sqrt{x}}\right) dx$

Solution:

$$\int \left(\sqrt{x} + \frac{1}{3\sqrt{x}}\right) dx = \int \sqrt{x} dx + \int \frac{1}{3\sqrt{x}} dx$$

$$= \int (x)^{1/2} dx + \frac{1}{3} \int (x)^{-1/2} dx$$

$$= \frac{(x)^{\frac{1}{2}+1}}{\frac{1}{2}+1} + \frac{1}{3} \cdot \frac{(x)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + c$$

$$= \frac{(x)^{\frac{3}{2}}}{\frac{3}{2}} + \frac{1}{3} \cdot \frac{(x)^{\frac{1}{2}}}{\frac{1}{2}} + c = \frac{2}{3} x^{\frac{3}{2}} + \frac{2}{3} x^{\frac{1}{2}} + c$$

8. $\int (e^x + 4e^{3x}) dx$

Solution:

$$\int (e^x + 4e^{3x}) dx = \int e^x dx + \int 4e^{3x} dx$$

$$=e^x + 4 \int e^{3x} dx = e^x + 4 \cdot \frac{e^{3x}}{3} + c = e^x + \frac{4}{3}e^{3x} + c$$

$$9. \int (6^x + x^2 + e^x) dx$$

Solution:

$$\begin{aligned} \int (6^x + x^2 + e^x) dx &= \int 6^x dx + \int x^2 dx + \int e^x dx \\ &= \frac{6^x}{\log 6} + \frac{x^3}{3} + e^x + c \end{aligned}$$

$$10. \int 4 \sin\left(\frac{\theta}{3}\right) d\theta$$

Solution:

$$\begin{aligned} \int 4 \sin\left(\frac{\theta}{3}\right) d\theta &= 4 \int \sin\left(\frac{\theta}{3}\right) d\theta = 4 \cdot \frac{-\cos\left(\frac{\theta}{3}\right)}{\frac{1}{3}} + c \\ &= -12 \cos\left(\frac{\theta}{3}\right) + c \end{aligned}$$

$$11. \int \frac{\sin x + \cos x}{\cos x} dx$$

Solution:

$$\begin{aligned} \int \frac{\sin x + \cos x}{\cos x} dx &= \int \left(\frac{\sin x}{\cos x} + \frac{\cos x}{\cos x} \right) dx = \int \tan x dx + \int dx \\ &= \log(\sec x) + x + c \end{aligned}$$

$$12. \int \frac{1}{4+x^2} dx$$

Solution:

$$\int \frac{1}{4+x^2} dx = \int \frac{1}{2^2+x^2} dx = \frac{1}{2} \tan^{-1} \frac{x}{2} + c$$

$$13. \int \sin 2x \cos 3x dx$$

Solution:

$$\text{We know } \sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\int \sin 2x \cos 3x dx = \int \frac{1}{2} [\sin 5x + \sin(-x)] dx$$

$$= \frac{1}{2} \int \sin 5x dx - \frac{1}{2} \int \sin x dx = \frac{1}{2} \cdot \left(-\frac{\cos 5x}{5} \right) - \frac{1}{2} \cdot (-\cos x) + c$$

$$= -\frac{1}{10} \cos 5x + \frac{1}{2} \cos x + c$$

$$14. \int \frac{\sin x}{\cos^2 x} dx$$

Solution:

$$\int \frac{\sin x}{\cos^2 x} dx = \int \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} dx = \int \tan x \sec x dx = \sec x + c$$

15. $\int \cos^3 x dx$

Solution:

We know

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\therefore \cos^3 A = \frac{1}{4} [\cos 3A + 3 \cos A]$$

$$\begin{aligned} \int \cos^3 x dx &= \int \frac{1}{4} [3 \cos x + \cos 3x] dx \\ &= \frac{1}{4} \left[3 \int \cos x dx + \int \cos 3x dx \right] \\ &= \frac{1}{4} \left[3 \sin x + \frac{\sin 3x}{3} \right] + c \end{aligned}$$

Note:

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

Unit 3 Homework - 1

Evaluate the following integrals

1. $\int x^7 dx$

2. $\int \frac{1}{\sqrt{x}} dx$

3. $\int \sqrt[3]{x} dx$

4. $\int \frac{\cos x}{\sin^2 x} dx$

5. $\int \frac{dx}{\sin^2 x}$

6. $\int 4^x dx$

7. $\int \frac{x^2+1}{x^3} dx$

8. $\int \left(\frac{4e^{5x} - 9e^{4x} - 3}{e^{3x}} \right) dx$

Methods of integration

- integration by parts

Integration by parts

This method is used for integration of the product of two functions.

Let $f(x)$ and $g(x)$ be two functions (or u and v be functions of x) then

$$\begin{aligned}\int u(x)v(x)dx \\ = u(x) \cdot \int v(x)dx - \int \left[\frac{d}{dx}(u(x)) \int v(x)dx \right] dx\end{aligned}$$

In words,

$$\begin{aligned}\int (I^{st} function \times II^{nd} function) dx = \\ I^{st} function \int (II^{nd} function) dx \\ - \int \left(\frac{d}{dx} (I^{st} function) \times \int (II^{nd} function) dx \right) dx\end{aligned}$$

“The integral of the product of two functions = (first function) $\times$ (integral of the second function) – Integral of [(differential coefficient of the first function) $\times$ (integral of the second function)]”

$$\int I \cdot II = I \int II - \int (I' \int II)$$

Note:

- Here the important factor is the choice of I and II function in the product of two functions because either can be I or II function.
- The first function is to be chosen such that it reduces to a next lower term or to a constant term after subsequent differentiations. We adopt the following order

Logarithmic

Inverse Trigonometric

Algebraic

Trigonometric

Exponential

	I function	II function
1. $\int x \cos x \, dx$	x (being algebraic)	$\cos x$
2. $\int x^2 e^x \, dx$	x^2 (being algebraic)	e^x
3. $\int x^2 \log x \, dx$	$\log x$	x^2
4. $\int \frac{\log x}{(1+x^2)} \, dx$	$\log x$	$\frac{1}{(1+x)^2}$
5. $\int x \sin^{-1} x \, dx$	$\sin^{-1} x$	x
6. $\int \log x \, dx$	$\log x$	1

(In single function of logarithm and inverse trigonometric we take unity as II function)

- The formula of integration by parts can be repeated if necessary.
- 1 is sometimes taken as one of the factors and chosen as the second function.
- Sometimes the integral on the right hand side comes out to be the same as on the left hand side, then this integral is taken to the left hand side and the value of the integral is found.

Problems

Evaluate the following

1. $\int x \sin x \, dx$

Solution:

Integrating by parts we get

$$\begin{aligned} \int x \sin x \, dx & \quad I \quad \Pi \\ &= x \int \sin x \, dx - \int \left(\frac{d}{dx}(x) \cdot \int \sin x \, dx \right) dx \\ &= x(-\cos x) - \int 1 \cdot (-\cos x) dx \\ &= -x \cos x + \int \cos x \, dx + c = -x \cos x + \sin x + c \end{aligned}$$

2. $\int x e^x \, dx$

Solution:

Integrating by parts we get (choose x to be the first function)

$$\begin{aligned} \int x e^x \, dx & \quad I \quad \Pi \\ &= x \int e^x \, dx - \int \left(\frac{d}{dx}(x) \cdot \int e^x \, dx \right) dx \\ &= x e^x - \int 1 \cdot e^x \, dx = x e^x - \int e^x \, dx + c \\ &= x e^x - e^x + c \end{aligned}$$

3. $\int \log x \, dx$

Solution:

$$\begin{aligned} \int \log x \, dx &= \int 1 \cdot \log x \, dx = \log x \int 1 \cdot dx - \int \left(\frac{d}{dx}(\log x) \cdot \int 1 \cdot dx \right) dx \\ &= \log x \cdot x - \int \frac{1}{x} x \, dx = x \log x - x + c = x(\log x - 1) + c \end{aligned}$$

4. $\int e^x \sin x \, dx$

Solution:

$$\text{Let } I = \int e^x \sin x \, dx$$

$$I = \sin x \int e^x dx - \int \left(\frac{d}{dx} (\sin x) \int e^x dx \right) dx$$

$$I = \sin x \int e^x dx - \int \cos x e^x dx$$

$$I = \sin x e^x - \int \cos x e^x dx \\ = \sin x e^x - \left[\cos x \int e^x dx - \int \left(\frac{d}{dx} (\cos x) \int e^x dx \right) dx \right]$$

$$I = \sin x e^x - \left[\cos x e^x + \int \sin x e^x dx \right]$$

$$I = \sin x e^x - \cos x e^x - I + c$$

$$2I = \sin x e^x - \cos x e^x + c$$

$$I = \frac{1}{2} [\sin x e^x - \cos x e^x] + c$$

Note. Rule to evaluate $\int e^{ax} \sin bx \, dx$ or $\int e^{ax} \cos bx \, dx$

$$\text{Let } I = \int \overset{\text{I}}{e^{ax}} \overset{\text{II}}{\sin bx} \, dx \text{ or } \int \overset{\text{I}}{e^{ax}} \overset{\text{II}}{\cos bx} \, dx$$

Integrate twice by product Rule and transpose term containing I from R.H.S. to L.H.S.

5. $\int x \log 2x \, dx$

Solution:

$$\int \overset{\text{I}}{x \log 2x} \, dx = \int (\log 2x) \cdot \overset{\text{II}}{x} \, dx$$

$$\text{Applying Product Rule } I \int II \, dx - \int \left(\frac{d}{dx} (I) \int II \, dx \right) dx$$

$$= (\log 2x) \int x \, dx - \int \left(\frac{d}{dx} (\log 2x) \int x \, dx \right) dx$$

$$= (\log 2x) \frac{x^2}{2} - \int \frac{1}{2x} \cdot 2 \cdot \frac{x^2}{2} \, dx$$

$$\begin{aligned}
 &= \frac{1}{2} x^2 \log 2x - \frac{1}{2} \int x \, dx \quad \left[\because \frac{x^2}{x} = \frac{x \cdot x}{x} = x \right] \\
 &= \frac{1}{2} x^2 \log 2x - \frac{1}{2} \frac{x^2}{2} + c = \frac{x^2}{2} \log 2x - \frac{x^2}{4} + c.
 \end{aligned}$$

6. $\int x^2 \log x \, dx$

Solution:

$$\int \underset{\text{I}}{x^2} \log \underset{\text{II}}{x} \, dx = \int (\log x) x^2 \, dx$$

Applying Product Rule: $\int \text{I} \int \text{II} \, dx = \int \left(\frac{d}{dx} (\text{I}) \int \text{II} \, dx \right) dx$

$$= \log x \int x^2 \, dx - \int \left(\frac{d}{dx} (\log x) \int x^2 \, dx \right) dx$$

$$\begin{aligned}
 &= (\log x) \frac{x^3}{3} - \int \frac{1}{x} \frac{x^3}{3} \, dx = \frac{x^3}{3} \log x - \frac{1}{3} \int x^2 \, dx \quad \left[\because \frac{x^3}{x} = x^2 \right] \\
 &= \frac{x^3}{3} \log x - \frac{1}{3} \frac{x^3}{3} + c = \frac{x^3}{3} \log x - \frac{x^3}{9} + c.
 \end{aligned}$$

7. $\int x \tan^{-1} x \, dx$

Solution:

$$\begin{aligned}
 \text{Let } \text{I} &= \int x \tan^{-1} x \, dx = \int (\tan^{-1} x) \cdot x \, dx \\
 &= (\tan^{-1} x) \cdot \frac{x^2}{2} - \int \frac{1}{1+x^2} \cdot \frac{x^2}{2} \, dx
 \end{aligned}$$

$$\begin{aligned}
&= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx \\
&= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2} \right) dx \\
&\quad \left[\because \frac{x^2}{1+x^2} = \frac{1+x^2-1}{1+x^2} = 1 - \frac{1}{1+x^2} \right] \\
&= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} (x - \tan^{-1} x) + c \\
&= \frac{1}{2} [x^2 \tan^{-1} x - x + \tan^{-1} x] + c = \frac{1}{2} [(x^2 + 1) \tan^{-1} x - x] + c.
\end{aligned}$$

8. $\int x \sin^{-1} x \, dx$

Solution:

$$\begin{aligned}
\int x \sin^{-1} x \, dx &= \sin^{-1} x \int x \, dx - \int \left(\frac{d}{dx} (\sin^{-1} x) \int x \, dx \right) dx \\
&= \sin^{-1} x \cdot \frac{x^2}{2} - \int \frac{1}{\sqrt{1-x^2}} \cdot \frac{x^2}{2} dx = \sin^{-1} x \cdot \frac{x^2}{2} - \frac{1}{2} \int \frac{x^2}{\sqrt{1-x^2}} dx \text{ -----(1)}
\end{aligned}$$

Consider $\int \frac{x^2}{\sqrt{1-x^2}} dx$

Let $x = \sin t \Rightarrow t = \sin^{-1} x$

Differentiating $x = \sin t$ w.r.t. x we get

$$\begin{aligned}
1 &= \cos t \cdot \frac{dt}{dx} \therefore dx = \cos t \, dt \therefore \int \frac{x^2}{\sqrt{1-x^2}} dx = \int \frac{\sin^2 t}{\sqrt{1-\sin^2 t}} \cos t \, dt \\
&= \int \frac{\sin^2 t}{\sqrt{\cos^2 t}} \cos t \, dt = \int \frac{\sin^2 t}{\cos t} \cos t \, dt = \int \sin^2 t \, dt = \int \frac{1-\cos 2t}{2} dt = \int \frac{1}{2} dt - \int \frac{\cos 2t}{2} dt \\
&= \frac{t}{2} - \frac{\sin 2t}{4} + C = \frac{\sin^{-1} x}{2} - \frac{2 \sin t \cos t}{4} + C \\
&= \frac{\sin^{-1} x}{2} - \frac{2 \sin(\sin^{-1} x) \sqrt{1-\sin^2 t}}{4} + C = \frac{\sin^{-1} x}{2} - \frac{2x\sqrt{1-x^2}}{4} + C
\end{aligned}$$

Substituting in (1) we get

$$\int x \sin^{-1} x \, dx = \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \left[\frac{\sin^{-1} x}{2} - \frac{2x\sqrt{1-x^2}}{4} \right] + c$$

Unit 3 Homework - 2

Evaluate the following integral

1. $\int \sec^2(2 + 3x) dx$
2. $\int \sec(3x + 5) \tan(3x + 5) dx$
3. $\int (x + 1)^4 dx$
4. $\int \frac{dx}{\sqrt{5-9x}}$
5. $\int e^{3-8x} dx$
6. $\int \cos 4x \cos 2x \, dx$
7. $\int x^2 \sin x \, dx$
8. $\int \sin^{-1} x \, dx$
9. $\int x \cos 2x \, dx$
10. $\int x^2 e^{3x} \, dx$
11. $\int x (\log x)^2 \, dx$